3.7.1 The collect layer: diverse data sources

The collect layer encompasses all the methods we use to capture information about the physical world. As figure 3.11 shows, this includes four distinct approaches, each addressing different constraints:

- *Electronic sensing*—sensors measuring rooms, appliances, and indoor plants capture environmental parameters directly. These sensors vary in complexity from simple temperature measurements to multi-parameter devices monitoring air quality, occupancy, and soil conditions.
- *Electronic sensors with edge processing*—the water meter camera represents sensing couple with edge processing where raw data (images) gets processed locally into actionable information (meter readings). This approach solves the bandwidth and power constraints that would make transmitting full images impractical over LoRaWAN.
- *Manual data collection*—the swimming pool represents measurements that are more cost-effective to collect manually than to automate. Pool chemistry testing with a $50 digital meter provides the same accuracy as a $1,000 automated system, making manual collection the pragmatic choice.
- *File-based integration*—the electricity meter shows how existing infrastructure often provides data in formats that require special handling. My optical pulse sensor captures detailed consumption data but only syncs to a mobile phone, requiring CSV export and manual upload to integrate with the digital twin.

This diversity reflects real-world constraints where cost, technology limitations, and practical considerations prevent a uniform approach to data collection.

3.7.2 The connect layer: protocol translation and routing

The connect layer handles the complex task of moving data from diverse sources to a common processing platform. Three distinct communication paths demonstrate how different technologies serve different requirements: